

Learning outcome 3: Configure CCTV System

A. Management of Administration parameters settings

A parameter is a specific configurable option within a system that controls how it behaves. Think of it as a control point you can adjust to customize the system.

In a CCTV system, parameters are the individual items you can modify inside the configuration menu.

◆ Examples of CCTV Parameters

- Default password
- Recording storage location
- Recording schedule
- Video resolution
- Date and time settings
- Network IP address
- Motion detection sensitivity

A **setting** is the actual value you assign to a parameter.

If a parameter is the option, the setting is the choice you make for that option.

1 Change Default Password

◆ What is a Default Password?

A default password is the original factory-set password that comes with a device when it is new.

For Hikvision devices:

- The default username is usually admin
- Older models used 12345
- Newer models require you to create a password during first activation

⚠ Why this matters:

- Default passwords are publicly known
- Leaving them unchanged makes the system vulnerable to hacking
- It is the first step in securing a CCTV system

🔒 Steps to Change Password on Hikvision Devices

A. On DVR (Digital Video Recorder)

Using Local Monitor (Mouse + Screen connected to DVR)

1. Log in using current admin password
2. Go to Menu
3. Click Configuration
4. Select User Management
5. Choose admin
6. Click Modify Password
7. Enter:
 - Old password
 - New strong password
 - Confirm password
8. Click Save / OK

B. On NVR (Network Video Recorder)

1. Log in
2. Go to Menu
3. Select Configuration
4. Open User or User Management
5. Select admin
6. Click Edit / Modify Password
7. Enter old and new password
8. Save

C. On IP Camera (Hikvision) – Method 1: Using Web Browser

1. Connect camera to network
2. Enter camera IP address in browser
3. Log in
4. Go to Configuration
5. Select User Management
6. Click Modify Password
7. Enter old password
8. Enter new password

2 Set Recording Storage

◆ What Does "Set Recording Storage" Mean?

It means choosing and configuring where video footage will be saved.

- Internal Hard Disk Drive (HDD)
- External HDD
- Network storage
- NAS

- ✓ Ensures video is saved properly
- ✓ Prevents "No Disk" errors
- ✓ Avoids data loss
- ✓ Allows continuous recording
- ✓ Enables playback of recorded footage

If storage is not configured:

- Cameras may show live view
- But recording will NOT happen

📁 Steps to Set Recording Storage on DVR

1. Log in
2. Go to Menu
3. Click Configuration
4. Select Storage
5. Open HDD Management
6. Check HDD status (Normal / Uninitialized)
7. If uninitialized:
 - Select HDD
 - Click Init
8. Save settings
9. Optional: Go to Record Schedule, Enable continuous or motion recording

📁 Steps to Set Recording Storage on NVR

1. Log in

2. Go to Menu
3. Select Configuration → Storage → HDD Management
4. Check HDD status
5. If new HDD, select and initialize
6. Set recording schedule under Record → Schedule

Management of CCTV User Accounts

User account management controls who can access the CCTV system and what they are allowed to do.

Proper management improves:

- Security
- Accountability
- System control
- Data protection

1 Identification of User Account

It is the process of recognizing and defining different users in the CCTV system. Each user must have:

- A unique username
- A password
- Assigned access rights (permissions)

◆ Common User Types in Hikvision Devices

User Type	Access Level
Admin	Full control of system
Operator	Live view + playback
Guest	Live view only

- ✓ Prevents unauthorized access
- ✓ Tracks user activities
- ✓ Limits system misuse
- ✓ Improves security control

2 Creation of User Account

It is the process of adding a new authorized person to access the CCTV system.

📁 Steps to Create User Account on DVR / NVR

1. Log in as admin
2. Go to Menu → Configuration → User Management
3. Click Add User
4. Enter username and password
5. Assign permissions:
 - Live view
 - Playback
 - PTZ control
 - Configuration rights
6. Click Save

ASSIGN PERMISSION EXPLAINED

1 Live View Permission

Allows the user to watch real-time video from connected cameras.

- View live camera footage
- Switch between cameras
- View multiple camera screens
- Monitor activities as they happen

Cannot: watch recorded footage, change settings, control PTZ (unless granted).

2 Playback Permission

Allows access to previously recorded video footage.

- Search recordings by date/time
- Replay past events
- Pause, fast-forward, rewind
- Download or back up footage (if allowed)

3 PTZ Control Permission

Allows control of Pan, Tilt, Zoom cameras.

- Pan, Tilt, Zoom, Set presets, Patrol
- Used for tracking moving objects

4 Configuration Rights

Allows changing system settings including cameras, storage, schedules, and user accounts.

⚠ Usually only the Admin should have full configuration rights.

🌐 On IP Camera (Web Interface)

1. Enter camera IP in browser
2. Log in as admin
3. Go to Configuration → User Management
4. Click Add, enter details
5. Assign permissions and Save

3 Setting of Factor Authentication

Authentication means verifying a user's identity before granting access.

🔑 Knowledge Factor Authentication

Something the user knows (password, PIN, security answer). Hikvision: username + password.

🔑 Two-Factor Authentication (2FA)

Requires two types of verification:

- Something you know (password)
- Something you have (phone / verification code)

In Hikvision: login password + code via Email or Hik-Connect.

Configure CCTV IP Addresses

IP addresses allow devices to communicate on a network.

1 Static IP Address

Manually assigned, does not change. Ensures stable remote access and port forwarding.

Example: NVR IP: 192.168.1.200, Subnet Mask: 255.255.255.0, Gateway: 192.168.1.1

Steps to Set Static IP on DVR / NVR

1. Log in as admin → Menu → Configuration → Network → General/TCP-IP
2. Uncheck DHCP
3. Enter IP, Subnet Mask, Gateway, DNS
4. Save & restart if required

2 Dynamic IP Address

Assigned automatically via DHCP. May change when device or router restarts. Not recommended for main DVR/NVR.

Steps to Enable Dynamic IP (DHCP)

1. Log in → Configuration → Network → TCP/IP
2. Check Enable DHCP
3. Save settings

Best Practice for CCTV Installations

- Use Static IP for DVR, NVR, main IP cameras
- Use Dynamic IP for temporary or small home systems

Apply Settings of CCTV Recordings

1 Video Format & Data Compression

Compression reduces video file size and network bandwidth usage.

Example: Raw 100GB → Compressed 10–20GB using H.264/H.265.

2 Encoding & 3 Decoding

Encoding compresses video for storage; decoding converts back for playback.

Steps to Set Video Format

1. Log in → Menu → Configuration → Video / Encoding
2. Select camera channel
3. Choose compression type (H.264 / H.265)
4. Save

2 Overwrite Capability

Automatically replaces old recordings when storage is full.

1. Menu → Configuration → Storage → Advanced/HDD settings
2. Enable Overwrite
3. Save

3 Backup Frequency

How often footage is copied for safekeeping (Daily, Weekly, Monthly, After incident).

1. Insert USB → Playback → Search → Select video → Export / Backup → Choose format → Start

4 Set Time and Date

1. Menu → Configuration → System → Time Settings
2. Set Date, Time, Time zone
3. Enable NTP if connected to internet
4. Save settings

5 Set Recording Mode

Defines how and when cameras record:

- Continuous Recording – 24/7
- Motion Detection Recording – only when motion detected
- Scheduled Recording – specific times

1. Menu → Configuration → Record → Schedule
2. Select camera channel
3. Select recording type (Continuous / Motion / Schedule)
4. Set time grid
5. Save settings

Documenting CCTV Functionalities

1 Introduction to Documentation

Provides a detailed record of the CCTV system, components, configuration, and usage.

- System overview
- Troubleshooting
- Legal compliance
- Guides upgrades

2 CCTV System Requirements

Defines what is needed for the CCTV to operate effectively:

- Hardware: DVR/NVR, Cameras, HDD, Power, Network
- Software: CMS, Mobile App, Firmware
- Environment: mounting, lighting, temperature/humidity

3 Tools, Materials, and Equipment

Category	Examples
Tools	Screwdrivers, crimping tools, cable testers
Materials	Cables, connectors, brackets
Equipment	Cameras, DVR/NVR, monitors, storage, power

4 CCTV Diagram

A CCTV diagram visually represents how the system components are connected. It provides a clear reference for installation, troubleshooting, and future expansion.

◆ Key Elements

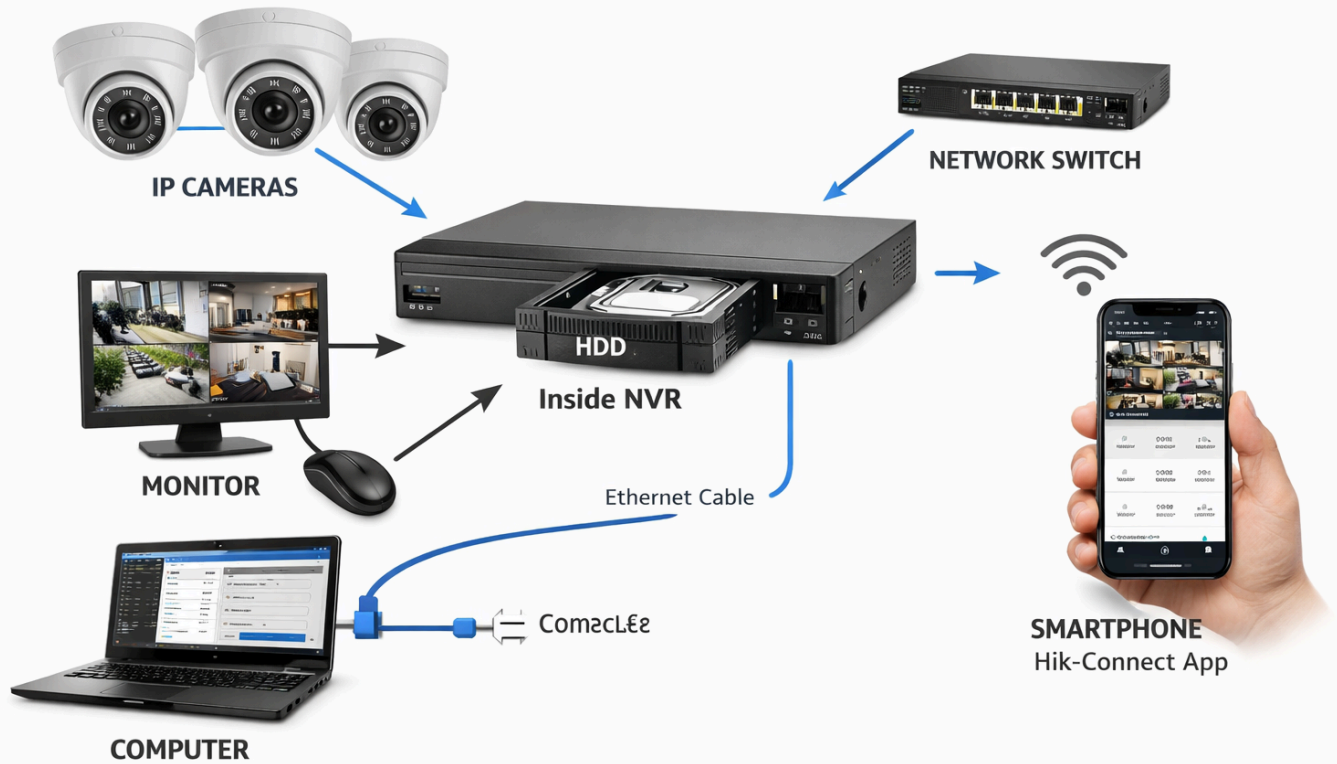
- IP Camera locations
- NVR position
- Monitor connected to NVR
- Mouse connected to NVR
- HDD inside NVR
- Network Switch connectivity
- Computer connected to NVR via Ethernet (with configuration software)
- Smartphone using Hik-Connect app
- Power sources for all devices

◆ Purpose

- Helps technicians understand the complete system layout
- Assists in installation, maintenance, and troubleshooting
- Supports future expansion plans
- Provides a clear reference for configuring remote access (computer & smartphone)

💡 **Visual Representation:** IP Cameras → NVR → Network Switch → Monitor / Mouse, with HDD inside NVR; Computer connected via Ethernet for configuration; Smartphone connected via Hik-Connect app.

CCTV SYSTEM CONNECTION



5 Cost Documentation

Definition: Cost documentation is the process of recording and estimating all expenses involved in setting up a CCTV system, including device specifications, unit costs, and total expenditure. It provides a clear reference for budgeting, planning, and accountability.

Cost estimation is made up of three components:

1. **Item Specifications** – Description or model of the device.
2. **Item Costs** – Price per unit in Rwandan Francs (RWF).
3. **Total Cost** – Sum of all item costs.

✦ 1. Item Specifications

Item	Specifications	Quantity
IP Camera	Hikvision DS-2CD2143G0-I, 4MP, IR	4
NVR	Hikvision DS-7204HQHI-K1, 4 Channel	1
Hard Drive	Western Digital Purple 2TB (inside NVR)	1
Network Switch	8-Port Gigabit Switch	1
Monitor	22" LED Monitor	1
Mouse	USB Mouse for NVR	1
Cables & Connectors	UTP Cat6 cables + BNC connectors	1 set

Computer	Laptop with configuration software	1
Smartphone	Mobile device with Hik-Connect App	1
Installation / Labor	Mounting, setup, configuration	1

 2. Item Costs

Item	Quantity	Unit Cost (RWF)	Total Cost per Item (RWF)
IP Camera	4	150,000	600,000
NVR	1	300,000	300,000
Hard Drive	1	120,000	120,000
Network Switch	1	80,000	80,000
Monitor	1	150,000	150,000
Mouse	1	15,000	15,000
Cables & Connectors	1 set	50,000	50,000
Computer	1	600,000	600,000
Smartphone	1	400,000	400,000
Installation / Labor	1	100,000	100,000

 3. Total Cost

Grand Total (RWF)	2,415,000
Maintenance / Contingency (RWF)	150,000
Final Total (RWF)	2,565,000

6 CCTV System Settings

Documenting system settings ensures consistency and easier troubleshooting:

- Network Configuration (IP addresses, gateway, subnet)
- Recording Settings (Video format, mode, overwrite, backup)
- User Accounts (permissions, authentication)
- Camera Settings (PTZ, resolution, motion sensitivity)
- Time & Date (timezone, NTP)